

- 1 Tree plantations are areas where trees are planted for a particular purpose. Some tree plantations increase the supply of wood for construction and fuel. In many plantations, fast-growing, alien tree species are planted.

Undergrowth is found in and around plantations, increasing plant biodiversity. Undergrowth is mainly made up of indigenous (native) small plants and shrubs.

Fig. 1.1 shows part of a plantation of eucalyptus trees.



Fig. 1.1

Bangladesh has many tree plantations as part of a national tree-planting programme.

Two fast-growing, alien tree species planted in Bangladesh are acacia, *Acacia auriculiformis*, and eucalyptus, *Eucalyptus camaldulensis*.

Acacia and eucalyptus are considered to be invasive species as they outcompete indigenous tree species such as sal tree, *Shorea robusta*, and mango, *Mangifera indica*.

Scientists carried out an investigation into the biodiversity of plant species in the undergrowth in plantations of alien trees compared with the undergrowth in plantations of indigenous trees.

The hypothesis that the scientists tested was stated as:

The undergrowth in plantations of alien tree species will have a lower biodiversity of plant species than the undergrowth in plantations of indigenous tree species.



(a) (i) Identify the independent variable in this investigation.

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For the investigation, plantations were selected within the same region of Bangladesh with the same environmental conditions.

- Three plantations of each tree species were selected: acacia, eucalyptus, sal tree and mango.
- A plot measuring 36 m by 36 m was studied within each plantation.
- For each plot, belt transects were used to collect the data needed to determine Simpson's index of diversity (D).

For each type of plantation, Simpson's index of diversity (D) was calculated for the plant species in the undergrowth.

Each plot was studied in April, July and November.

(ii) Identify **two** variables the scientists standardised in this investigation.

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- Describe how you could use belt transects to collect the data needed to determine Simpson's index of diversity (D) of the undergrowth in each of the plots.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



Table 1.1 shows the Simpson's index of diversity (D) values the scientists obtained for the different plantations during April, July and November.

Table 1.1

month	Simpson's index of diversity (D) for undergrowth plant species in plantation			
	acacia	eucalyptus	sal tree	mango
April	0.90	0.87	0.94	0.90
July	0.86	0.86	0.94	0.86
November	0.87	0.83	0.94	0.88
mean	0.88	0.85	0.94	0.88
$\pm s$	± 0.02	± 0.02	± 0.00	± 0.02

s = standard deviation

(c) State **and** explain whether the data in Table 1.1 support the hypothesis:

The undergrowth in plantations of alien tree species will have a lower biodiversity of plant species than the undergrowth in plantations of indigenous tree species.

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In China, a different group of scientists investigated whether the roots of invasive plant species produce chemicals that have an effect on the growth of other plants.

The scientists prepared an extract of the roots of staghorn sumac, *Rhus typhina*. Staghorn sumac is an invasive plant species.

The scientists prepared a stock solution of the root extract with a concentration of 10 mg cm^{-3} . The scientists then used the stock solution to prepare solutions with different concentrations of root extract.

(d) (i) Describe how the scientists could use the stock solution to prepare 50 cm^3 of solutions of the root extract with concentrations of 2.5 mg cm^{-3} and 7.5 mg cm^{-3} .

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In some regions of China, marigold plants, *Tagetes erecta*, are grown commercially.

In the investigation into the effect of staghorn sumac root extract on the growth of marigolds, the scientists carried out the procedure described in step 1 to step 5.

step 1 Prepare five containers, each with five marigold plants grown from seed.

step 2 Add 20cm³ of distilled water to one of the containers. This is the control.

step 3 Add 20cm³ of the root extract concentrations to the other four containers, so that each container has a different concentration of root extract.

step 4 Repeat step 2 and step 3 each day for 60 days.

step 5 After 60 days, remove the marigold plants from the soil and untangle the roots. Measure the length of the longest root for each plant.

The scientists replicated this procedure twice and calculated the mean maximum root length for the control and for the root extract concentrations.

The results are shown in Table 1.2.

Table 1.2

root extract concentration /mg cm ⁻³	mean maximum root length after 60 days /mm
control	163.0
2.5	153.2
5.0	156.4
7.5	152.0
10.0	141.6

- (ii) Using the data in Table 1.2, calculate the percentage change in the mean maximum root length between the control and the roots treated with 10mgcm⁻³ root extract concentration.

Show your working and record your answer to 3 significant figures.

percentage change [3]



- (e) A student suggested that this investigation could be improved.

With reference to the method used and the data in Table 1.2, state **and** explain **three** ways to improve confidence in the results.

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- (f) To investigate the effect of the different root extract concentrations on the diversity of the microorganisms in soil, the scientists:

- took samples of soil from each of the containers
- identified and counted the number of species of microorganism present in the samples
- calculated the Simpson's index of diversity (D) values for the species of microorganism in each of the soil samples.

State **two** variables the scientists should have standardised when taking the soil samples.

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[Total: 19]